

MSAI 699 Capstone Project

Capstone Project Proposal:

***NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence
Platform for Personalized Cognitive Health Education Using
Machine Learning and Retrieval-Augmented Generation***

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Abstract

AI techniques like natural language processing and large language models have greatly changed the way we obtain healthcare information (Bender et al., 2021; Lewis et al., 2020). However, general-purpose AI systems may provide us with unverified or incorrect data, which is a great danger when it comes to the field of cognitive health education where patients and caregivers require correct information about Alzheimer's disease, mild cognitive impairment, genetic factors and ways to change lifestyles (Alzheimer's Association, 2024; World Health Organization, 2023). The problem can be solved by NeuroGuide AI with its explainable multi-agent system using RAG, machine learning and explainability via SHAP technique for personalized and grounded cognitive health education and risk assessments (Breiman, 2001; Lundberg & Lee, 2017).

Proposal

Project Title

NeuroGuide AI: An Explainable Multi-Agent Artificial
Intelligence Platform for Personalized Cognitive Health Education
Using Machine Learning and Retrieval-Augmented Generation

Problem Statement, Motivation, and Impact

The introduction of large language models has brought a revolution in the way individuals are able to gather information related to healthcare; however, problems such as hallucinations, outdated data, and lack of transparency hinder user trust in this process and can even impact decision-making for healthcare (Bender et al., 2021). In particular, this is an extremely important issue in cognitive health education when users require correct information on Alzheimer's disease, mild cognitive impairment, genes, and other ways to influence one's health (Alzheimer's Association, 2024; World Health Organization, 2023).

In order to address this problem, the project proposes the development of *NeuroGuide AI* – a platform consisting of several agents working together, which will utilize the potential of Retrieval-Augmented Generation, machine learning, and explainable AI for providing trustworthy, personalized cognitive health education.

Project Scope

Due to the time constraint for this capstone being one semester, the development of the NeuroGuide AI system will not be a complete production-level system but a proof-of-concept prototype. This system will employ a five-agent architecture, which includes the Personalization, Retrieval, Reasoning, Validation, and Synthesis agents, together with a Retrieval-Augmented Generation pipeline and a machine learning–based educational risk assessment system using SHAP explainability. The primary goal of this capstone is to provide a proof of concept for implementing an evidence-based cognitive health education system using more than one technique of artificial intelligence. Further capabilities such as clinician-specific functionality and live clinical data feeds will be considered for the future.

Research Questions

Primary Research Question

Can an explainable multi-agent Retrieval-Augmented Generation (RAG) architecture provide more accurate, transparent, and trustworthy cognitive health education than a traditional single-agent conversational AI system?

Secondary Research Questions

1. **RQ1:** Does a multi-agent architecture improve the quality and reliability of educational responses compared to a single-agent approach?
2. **RQ2:** Can machine learning–based educational risk assessments improve personalization while maintaining transparency through explainable AI?
3. **RQ3:** How does Retrieval-Augmented Generation improve factual grounding and reduce unsupported responses?
4. **RQ4:** How do users perceive the usability, trustworthiness, and educational value of a multi-agent cognitive health platform?

Hypothesis

It is hypothesized that NeuroGuide AI will produce more accurate, transparent, and trustworthy educational responses than a traditional single-agent conversational AI system. By combining Retrieval-Augmented Generation, machine learning, explainable AI, and a multi-agent architecture, the platform is expected to improve factual grounding, reduce unsupported responses, and increase user confidence while providing personalized educational guidance.

Proposed AI Technologies

NeuroGuide AI integrates multiple artificial intelligence concepts explored throughout the Master of Science in Artificial Intelligence program.

Machine Learning

- Random Forest classification (Breiman, 2001)
- Educational risk prediction
- SHAP explainability

Natural Language Processing

- Large Language Models (GPT-4o-mini)
- TF-IDF document retrieval
- Latent Semantic Analysis (LSA)
- Prompt engineering

Retrieval-Augmented Generation

- Hybrid sparse and semantic document retrieval (Lewis et al., 2020)
- Citation-grounded response generation
- Confidence scoring

Multi-Agent Artificial Intelligence

The platform consists of five specialized AI agents (Wu et al., 2023)

1. Personalization
2. Retrieval
3. Reasoning
4. Validation
5. Response synthesis.

Human-Computer Interaction

- Personalized user interface
- Patient and clinician response modes
- Accessibility-focused interface design

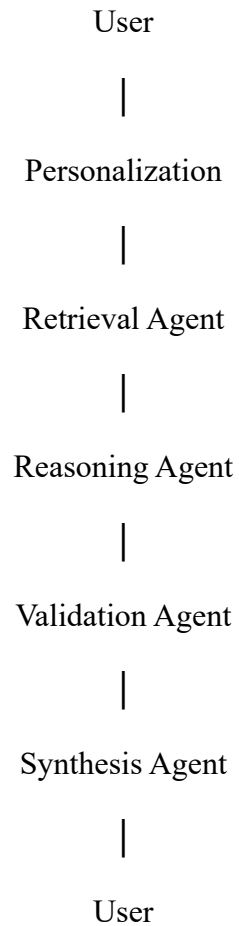
Responsible AI

- Explainable AI (SHAP)
- Bias awareness
- Source transparency
- Educational medical disclaimers

Proposed System Architecture

The NeuroGuide artificial intelligence system analyzes all user requests via a set of *five* different AI agents working together in harmony. In contrast to one language model being used, each particular agent is responsible for a specific task contributing to the high accuracy, integrity, and reliability of the final output.

Model architecture:



The proposed architecture consists of five collaborating AI agents:

Personalization

User profile information, including demographic and educational risk factors, is used to personalize educational responses and risk assessments.

Retrieval Agent

Searches trusted cognitive health resources and retrieves the most relevant supporting evidence.

Reasoning Agent

Analyzes retrieved information and generates evidence-based educational explanations tailored to the user's request.

Validation Agent

Verifies that generated responses are supported by retrieved evidence and assigns a confidence score to improve transparency.

Synthesis Agent

Produces the final educational response while adapting language for either patient or clinician audiences and providing supporting citations.

Datasets

The knowledge base will consist of trusted public sources related to Alzheimer's disease and cognitive health, including:

- PubMed Central
- NIH MedlinePlus (National Institute on Aging, 2024)
- Alzheimer's Association (Alzheimer's Association, 2024)
- Cochrane Reviews
- World Health Organization (WHO) (World Health Organization, 2023)
- Centers for Disease Control and Prevention (CDC, 2024)

The risk assessment model for education is constructed using publicly available dementia risk factors reported by the 2024 Lancet Commission (Livingston et al., 2024). These include demographic risk factors, lifestyle risk factors, cardiovascular risk factors, and APOE $\epsilon 4$ gene. The Random Forest Model will be trained using the Alzheimer's Prediction Dataset (Global), which is a publicly available dataset on Kaggle (Panday, 2025). This dataset includes labeled demographic, lifestyle, and clinical risk factors across several countries and is appropriate for supervised risk classification.

Evaluation

The effectiveness of NeuroGuide AI will be evaluated using quantitative and qualitative methods to assess system performance, machine learning accuracy, explainability, and user experience.

AI Performance

The Retrieval-Augmented Generation system will be evaluated using:

- Retrieval relevance
- Citation grounding
- Response confidence
- Response latency

Machine Learning

The educational risk assessment model will be evaluated using standard machine learning metrics, including:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Explainable AI

Model transparency will be assessed using SHAP feature attribution to verify that educational risk predictions are explainable and clinically reasonable (Lundberg & Lee, 2017).

Human-Computer Interaction

Considering the scope of this capstone, usability will be evaluated via heuristic evaluation and self-testing as opposed to external user tests. This will involve conducting heuristic evaluations based on established usability principles to identify interface improvements and assess overall user experience.

Ethical Considerations

NeuroGuide AI works within the field of healthcare education, which means that ethical AI use is crucial in this project (Bender et al., 2021). The system will operate only as an educational tool and won't give any diagnoses or suggestions on treatments. All answers will

contain disclaimers urging people to seek assistance from licensed healthcare providers regarding any medical concerns.

For additional transparency, all answers will contain citations of reliable medical sources wherever possible. Explanatory approaches such as SHAP will help users comprehend how particular risk factors influence educational risk assessments. Protected health information will not be collected and stored in the project. Possible sources of biases will be accounted for during the development of the project through using evidence-based medical resources and assessing the fairness of the answers.

Expected Challenges

There are some difficulties which are expected to occur while developing NeuroGuide AI. First, it would be necessary to make sure that the response generated by the AI is accurate and comes from trusted medical sources. It will be solved using the techniques such as Retrieval-Augmented Generation, source citations and validation agent.

Another difficulty may be related to finding balance between personalization and transparency. Explainable AI using SHAP will allow users to better understand how each factor affects the result while emphasizing on the fact that the platform is purely an educational one. The last difficulty will be integration of several AI elements into a user-friendly design.

Conclusion

The NeuroGuide AI system is an example of how a number of artificial intelligence approaches can be used together on one educational platform to generate reliable information

related to cognitive health care. The use of machine learning, explainable AI, Retrieval-Augmented Generation, and a multi-agent system aims to increase the efficiency and personalization of AI-based health care education in addition to demonstrating knowledge gained during the MSAI program. (Breiman, 2001; Lewis et al., 2020; Lundberg & Lee, 2017).

AI Disclosure: Artificial intelligence was used only to assist with proofreading, grammar refinement, formatting, and organization. The project idea, research direction, system design, implementation, technical decisions, analysis, and all final content were developed independently by the author, who assumes full responsibility for this submission.

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